# Purchase Prediction

# Based on Social Network Advertisements

Presenting By : G Akshaya Prabha
Roll No: 24100010029

# Aim

To develop a machine learning model to predict whether a user will purchase a product based on social network advertisement data.

# Objectives

- To analyze and understand customer data from social network advertisements
- To develop a machine learning model for predicting purchase behavior
- To evaluate the model using appropriate performance metrics
- To identify the key factors influencing customer purchasing decisions
- To design a user-friendly web interface for real-time prediction

# Introduction

This project uses machine learning to predict customer purchase decisions based on demographic and financial data, enabling better targeting and efficient marketing strategies.

## Advantages of the Model

- Helps in targeted advertising by identifying likely customers
- Improves marketing efficiency and reduces unnecessary costs
- Supports data-driven decision making
- Provides quick and real-time predictions through a web interface
- Identifies key factors influencing customer purchase behavior

# Procedure

- **Data Preprocessing:**

The dataset was cleaned and prepared by selecting relevant features such as Gender, Age, and Estimated Salary.

- **Encoding Categorical Data:**

The Gender feature was converted into numerical form to make it suitable for the machine learning model.

- **Train-Test Split:**

The dataset was divided into training and testing sets to evaluate the model's performance on unseen data.

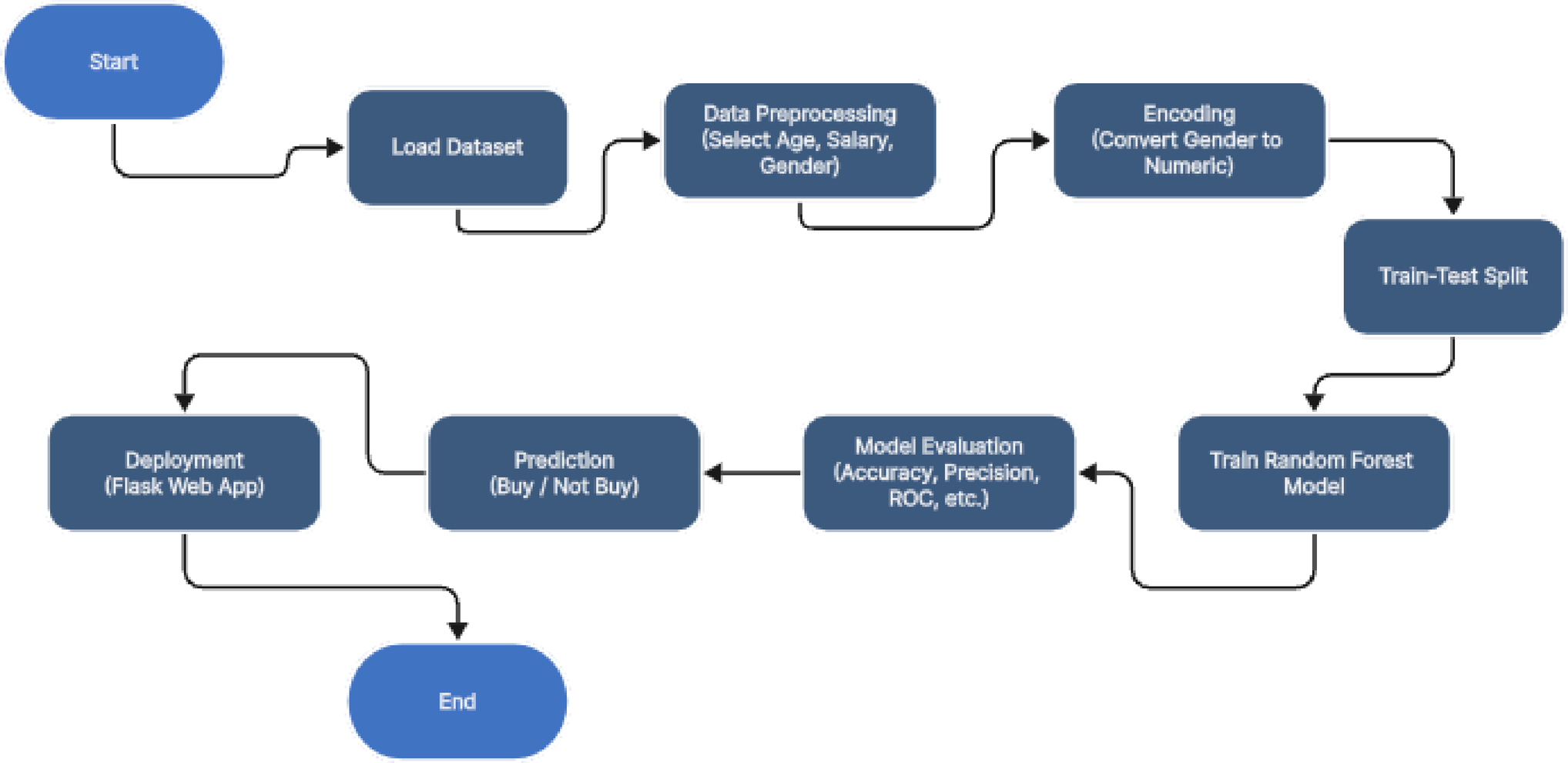
- **Model Training (Random Forest):**

A Random Forest Classifier was trained on the training data to learn patterns and relationships between features and customer purchase behavior.

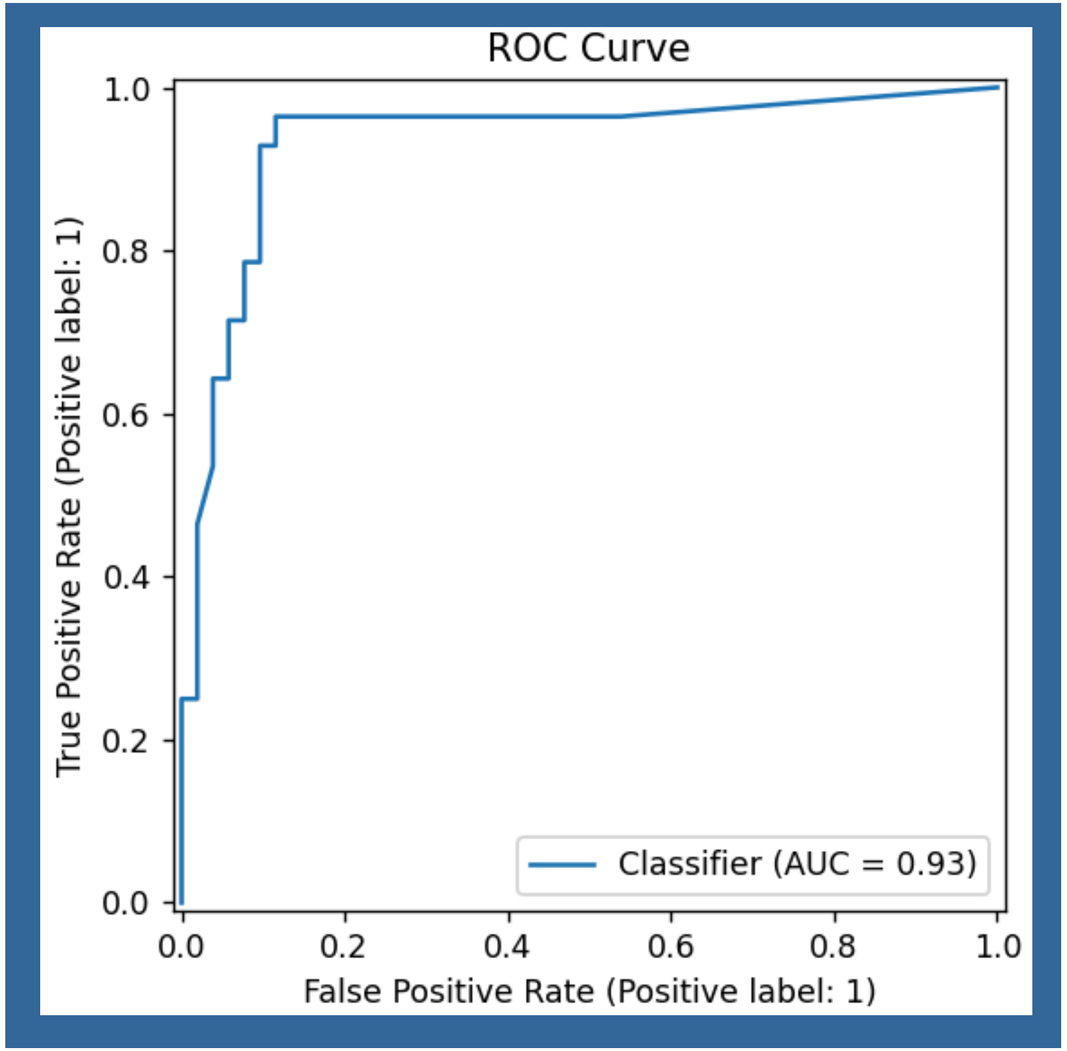
- **Prediction:**

The trained model was used to predict whether a customer will buy or not based on new input data.

# Model Workflow



# ROC-AUC Graph




**Axes of the ROC Curve**

**X-axis** → False Positive Rate (FPR)

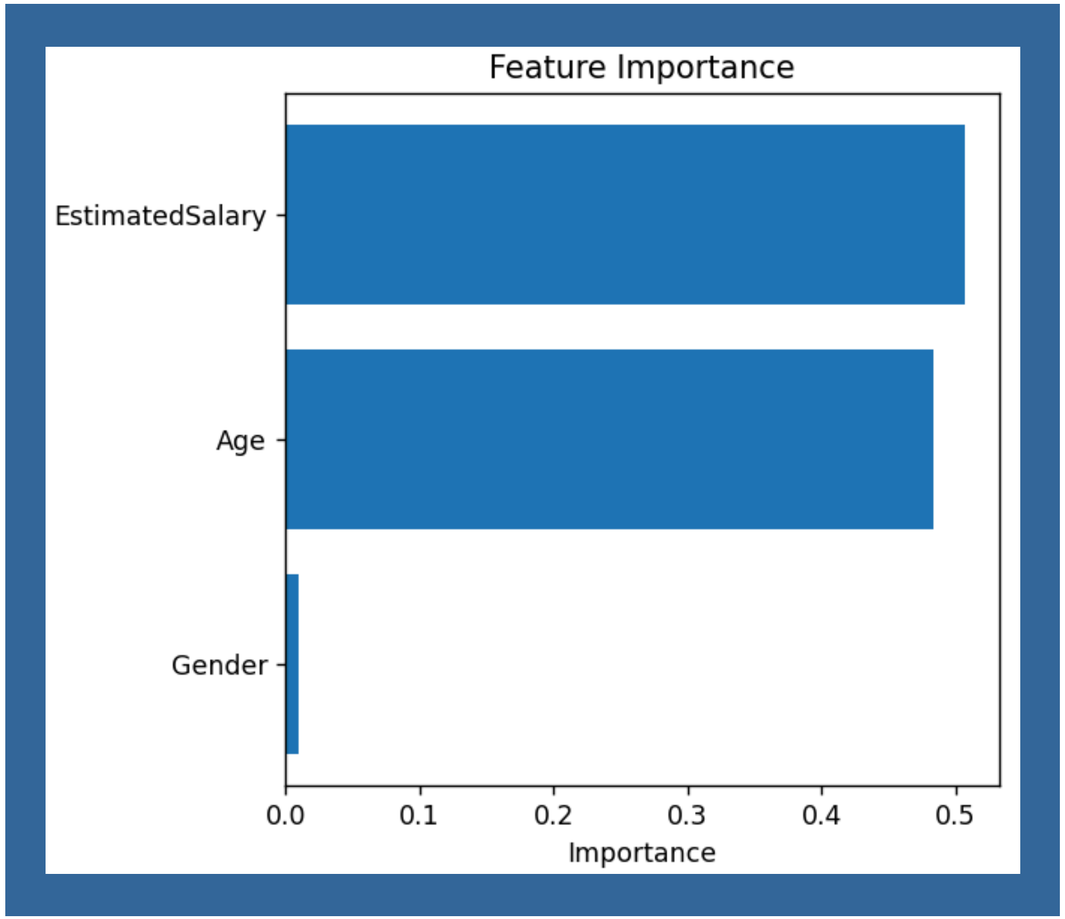
Non-buyers incorrectly predicted as buyers

**Y-axis** → True Positive Rate (TPR / Recall)

Actual buyers correctly predicted

**ROC curve is close to the top-left corner :**
Model is performing well

# Feature Importance

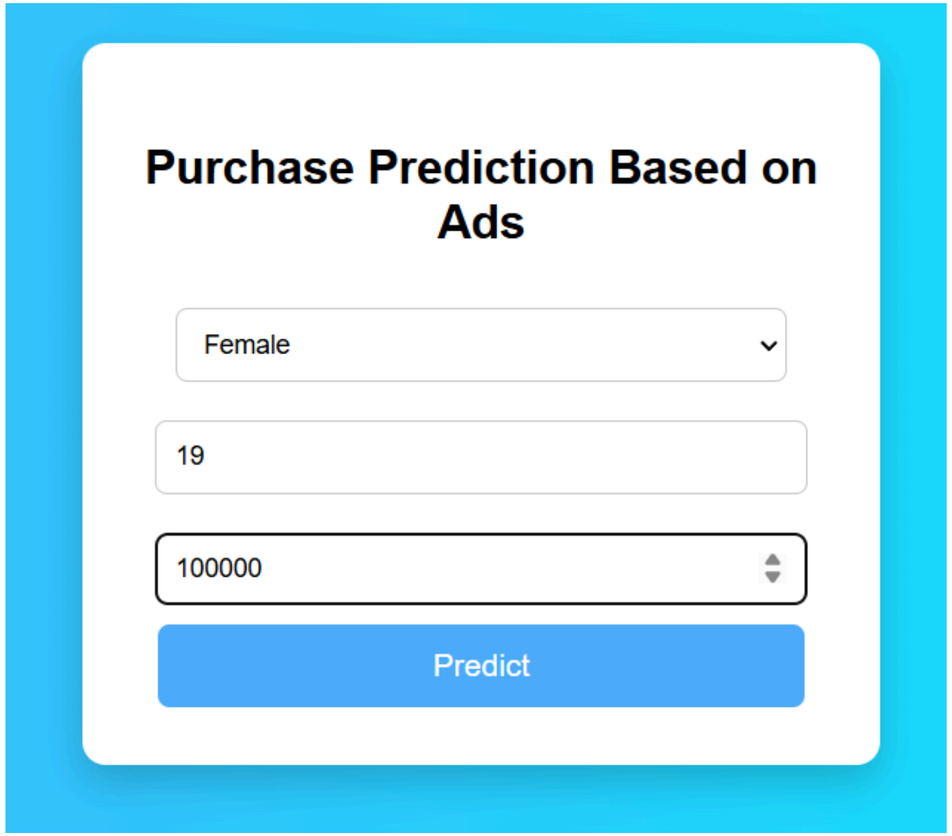


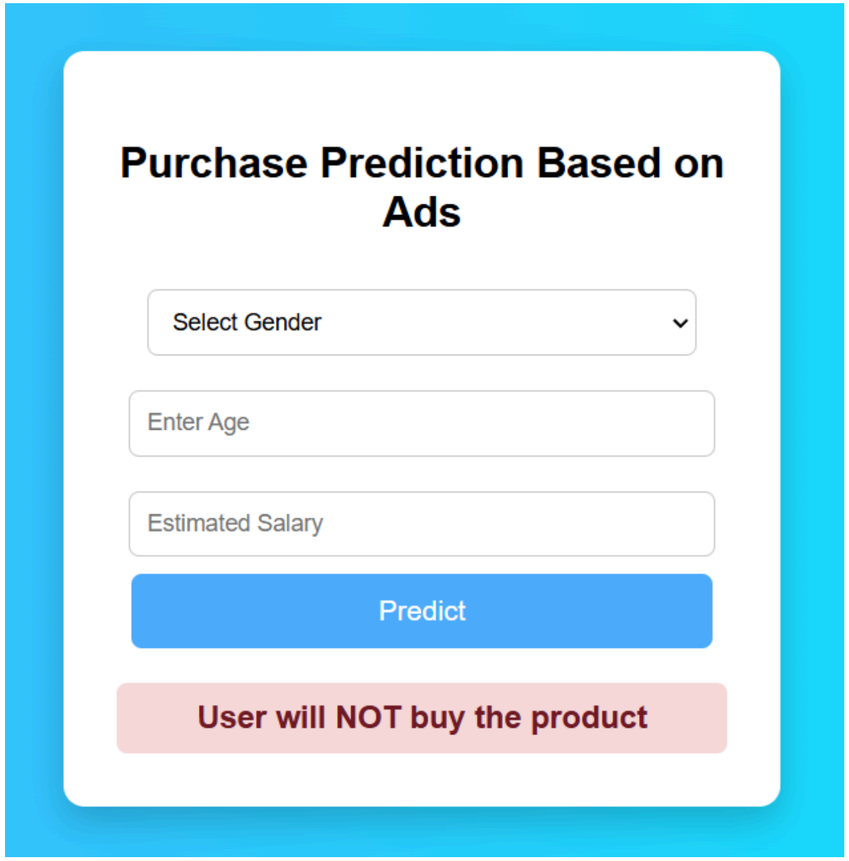

- **Estimated Salary (~0.51)**
  Most important feature with the highest impact on prediction. Customers with higher salaries are more likely to buy.

- **Age (~0.48)**
  Second most important feature and very close to salary in influence. Older customers have a higher chance of purchasing.

- **Gender**
  Very low importance and has minimal effect on prediction. The model largely ignores gender while making decisions.

# User Interface



# User Interface

**Input Fields :**

1. **Gender Dropdown**
- User selects Male or Female
- This input is used as one of the features for prediction
2. **Age Input**
- User enters their age
- Important factor influencing prediction
3. **Estimated Salary Input**
- User enters their salary
- One of the most important features in the model

**Predict Button :**

- When clicked, it sends user data to the backend (Flask)
- The machine learning model processes the input
- Returns the prediction result
- **Example shown:** “User will NOT buy the product”

**Simple flow :**

User Input → Click Predict → Data sent to model → Prediction generated → Result displayed